

第24讲. 场论初步

作业: P190. 12. 13 ①. 14.

一. 空间曲线积分与积分路径无关的条件.

定义1. $\Omega \subset \mathbb{R}^3$ 有界区域. 若 Ω 中任一条连续可微封闭曲线都可连续收缩为 Ω 中一点.称 Ω 是单连通.定理1. 设 $\Omega \subset \mathbb{R}^3$ 单连通区域. $P(x, y, z), Q(x, y, z), R(x, y, z) \in C^1(\Omega)$. 则下列等价:① $\int_{L(A, B)} Pdx + Qdy + Rdz$ 与积分路径无关. 只与起点 A 与终点 B 有关.② $\exists u(x, y, z) \in C^1(\Omega)$. s.t. $du = Pdx + Qdy + Rdz$.此时称 $u(x, y, z)$ 为 $Pdx + Qdy + Rdz$ 原函数.③ $\forall (x, y, z) \in \Omega$. $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial z}, \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}, \quad \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ ④ 对 Ω 内任意光滑或分段光滑封闭曲线 L , $\oint_L Pdx + Qdy + Rdz = 0$.定理2. 设 $\Omega \subset \mathbb{R}^3$ 有界区域. $P, Q, R \in C^1(\Omega)$. $\frac{1}{H} \exists u(x, y, z) \in C^1(\Omega)$ s.t. $du = Pdx + Qdy + Rdz$. 则 $\forall A, B \in \Omega$.

$$\int_{L(A, B)} Pdx + Qdy + Rdz = u(B) - u(A).$$

二. 场论初步.

数量场: $u(x, y, z)$. Ω .向量场. $\vec{F}(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z))$

1. 梯度场:

 $\Omega \subset \mathbb{R}^3$. $u(x, y, z)$ 为 Ω 上数量场. $u \in C^1(\Omega)$.则 $\forall M \in \Omega$. $\text{grad} u(M) = \left(\frac{\partial u}{\partial x}(M), \frac{\partial u}{\partial y}(M), \frac{\partial u}{\partial z}(M) \right)$

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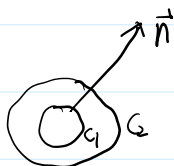
 $\nabla u(M)$ $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$ $\nabla u = \text{grad} u$

若 $\exists u \in C^1(\Omega)$, s.t. $\vec{F}(M) = \nabla u(M)$, 称 $\vec{F}$ 为有势场. 称 u 为 $\vec{F}$ 势函数.
($\forall M \in \Omega$)

$$u(x, y, z) \in C^1(\Omega), \quad \forall M \in \Omega, \quad \nabla u(M) \neq 0.$$

$\forall M \in \left\{ (x, y, z) \mid u(x, y, z) = C \right\}$ — 等值面.

$$\vec{n} = \nabla u(M)$$



$$C_1 < C_2.$$

2. 散度场.

定义2. 设 $\vec{F} = (P, Q, R) \in C^1(\Omega)$, $\forall M \in \Omega$.

称 $\text{div } \vec{F}(M) = \underline{P'_x(M) + Q'_y(M) + R'_z(M)}$ 为 $\vec{F}$ 在 M 散度.

定义3. 设 $\vec{F} = (P, Q, R) \in C^1(\Omega)$, $\Sigma \subset \Omega$ 为一片定向曲面, $\vec{n}$ 为 Σ^+ 单位法向量.

称 $\iint_{\Sigma^+} \vec{F} \cdot \vec{n} \, dS = \iint_{\Sigma^+} P \, dy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy$ 为 $\vec{F}$ 通过 Σ^+ 通量.

$\vec{F} = (P, Q, R) \in C^1$
流场.

以 $\forall M_0 \in \Omega$ 为圆心, $\varepsilon > 0$ 为半径作球 $B_\varepsilon \subset \Omega$.

$$\oint_{\partial B_\varepsilon} \vec{F} \cdot \vec{n} \, dS = \iiint_{B_\varepsilon} \text{div } \vec{F} \, dV \quad (\text{Gauss公式})$$

$$\underline{\underline{\exists \vec{z} \in B_\varepsilon \quad \text{div } \vec{F}(\vec{z}) \, V_{B_\varepsilon}}}$$

$$\text{div } \vec{F}(\vec{z}) = \frac{1}{V_{B_\varepsilon}} \oint_{\partial B_\varepsilon} \vec{F} \cdot \vec{n} \, dS \quad \text{— 通量体密度}$$

$$\varepsilon \rightarrow 0^+ \Rightarrow \vec{z} \rightarrow M_0.$$

$$\text{div } \vec{F}(M_0) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{V_{B_\varepsilon}} \oint_{\partial B_\varepsilon} \vec{F} \cdot \vec{n} \, dS,$$

$\text{div } \vec{F}(M_0) \neq 0$. 流场 $\vec{F}$ 在 M_0 有流源.

若 $\text{div} \vec{F}(M_0) > 0$, ... 有正流源, 称 M_0 为源点.

$\text{div} \vec{F}(M_0) < 0$, ... 负 ... 称 M_0 为汇点.

$\text{div} \vec{F}(M_0) = 0$,

定义3. 若 $\vec{F}(M) = (P(M), Q(M), R(M)) \in C^1(\Omega)$, $\forall M \in \Omega$, $\text{div} \vec{F}(M) = 0$.

则称 $\vec{F}$ 为 Ω 上的无源场.

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

$$\Delta = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \text{--- Laplace 算符.}$$

称 $\Delta u = 0$ 为 Laplace 方程.

若 $u \in C^2(\Omega)$ 且 $\Delta u = 0$, 称 u 为调和函数.

定义4. 设 $u(x, y, z) \in C^2(\Omega)$ 是调和函数. 称 ∇u 为调和场.

练习: 设 $\vec{F} = (P, Q, R) \in C^1(\Omega)$ 是有势场, 又是无源场, 则 $\vec{F}$ 调和场.

3. 旋度场.

定义5. 设 $\vec{F} = (P, Q, R) \in C^1(\Omega)$, $\forall M \in \Omega$.

$$\text{rot} \vec{F}(M) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$

$\downarrow$
 $\vec{F}$ 在 M 旋度

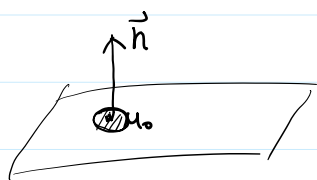
$$\left\| \begin{array}{ccc} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{array} \right\|_M = \nabla \times \vec{F}(M)$$

定义6. 设 $\vec{F} = (P, Q, R) \in C(\Omega)$, $L \subset \Omega$ 一条光滑或分段光滑定向曲线.

$\vec{\tau}$ 为 L^+ 单位切向量. 称 $\oint_{L^+} \vec{F} \cdot \vec{\tau} ds = \oint_{L^+} P dx + Q dy + R dz$ 为 $\vec{F}$ 沿 L^+ 环量.

$\vec{F} = (P, Q, R) \in C^1(\Omega)$. 流场. $\forall M_0 \in \Omega$. $\forall \vec{n}$. $\|\vec{n}\| = 1$.

$\vec{n}$



D_r : M_0 为中心. $r > 0$ 半径. $D_r \subset \Omega$.

$$\oint_{(\partial D_r)^+} \vec{F} \cdot \vec{c} \, ds = \iint_{(D_r)^+} \underbrace{\text{rot } \vec{F} \cdot \vec{n}}_{\substack{\exists \eta \in D_r \\ \text{rot } \vec{F}(\eta) \cdot \vec{n}}} \, dS \quad (\text{Stokes ad'})$$

$$\text{rot } \vec{F}(\eta) \cdot \vec{n} = \frac{1}{S_{D_r}} \oint_{(\partial D_r)^+} \vec{F} \cdot \vec{c} \, ds. \quad \text{—— 环量面密度.}$$

$$r \rightarrow 0 \Rightarrow \eta \rightarrow M_0.$$

$$\underbrace{\text{rot } \vec{F}(M_0) \cdot \vec{n}}_{\substack{\downarrow \\ \vec{F} \text{ 在 } M_0 \text{ 环绕 } \vec{n} \text{ 方向旋量}}} = \lim_{r \rightarrow 0} \frac{1}{S_{D_r}} \oint_{(\partial D_r)^+} \vec{F} \cdot \vec{c} \, ds.$$

若 $\text{rot } \vec{F}(M_0) \neq 0$. 称 M_0 为 $\vec{F}$ 的漩涡.

$\|\text{rot } \vec{F}(M_0)\|$ 越大. $\vec{F}$ 在 M_0 环绕 $\vec{n}$ 旋转越大.

$$\text{rot } \vec{F}(M_0) = 0.$$

$$\text{rot } \vec{F}(M_0)$$